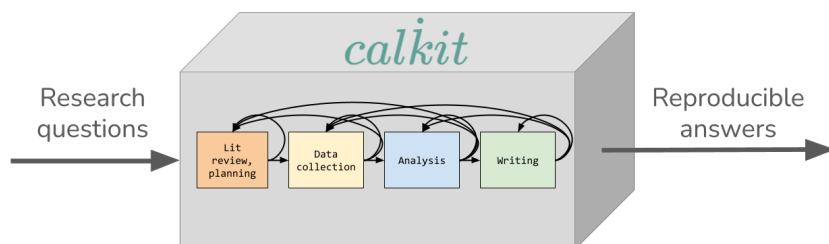


Calkit: Continuously integrated, transparent, reproducible science in the age of agentic AI



BY THE NUMBERS

1.3× as many papers in a PhD with agents alone
2.2× as many with agents and Calkit

Simulated in Calkit's own pipeline.

Generative AI is making it much cheaper to produce research outputs, while the reproducibility crisis remains unsolved (**only 12% of studies share code**, and even fewer share code that is quick to verify). As the bottleneck is shifting from producing outputs to checking that those outputs are trustworthy, it's now more important than ever to make it intuitive and enjoyable to work transparently and reproducibly, both for humans and AI agents.

Rather than assuming we should simply teach scientists software development tools and platforms and leave them to assemble their own workflows (the status quo, which seems to have saturated), Calkit provides a vertically-integrated, research-focused experience right out of the box: reference management, version control (including large data files), computational environments, pipeline automation, and a web-based, self-hostable collaboration platform.

Calkit is designed to help individuals work more efficiently by allowing all stages of research—from asking questions, to data collection and analysis, to publication—to live together in the same project, breaking down silos that slow down iteration while allowing more inclusive collaboration, i.e., from those who don't want to or need to learn complex tools like Git. A Calkit project includes the research article, not only supplements it, and is intended to be shipped whole, providing full context and lighting fast verifiability, by linking all evidence for answers to research questions back to reproducible calculations.

A success story

Rebecca McCabe recently finished her PhD in Mechanical Engineering at Cornell. She used Calkit to create a single-button reproducible pipeline that integrated MATLAB, Python, Julia, LaTeX, and more. She used Calkit's Overleaf integration to allow collaborators to contribute without learning Git.

Where we're heading and what support we need

Calkit is developed and operated in a very lean and efficient manner. We're currently supported with 25% of a Sr. Research Software Engineer's capacity provided by the Schmidt Academy for Software Engineering at Caltech. Google Cloud has generously provided credits to run the cloud infrastructure for a year, and we need \$5k to stay running for another.

With an MVP in place serving a handful of users, we now need to do some training and community building by hosting workshops, starting around Southern California and Boston. These will be focused on computational reproducibility principles, with Calkit positioned as one, not the only way to achieve them. Through these efforts we're looking to 10x our user base and continue to evolve the software to increase the number of comprehensive, transparent, single-button reproducible research projects released alongside papers. To do this, we're seeking \$75k for 2027.

Learn more

To learn more about Calkit, you can check out [the docs](#) and try out the web platform at calkit.io.